

CRUDE OIL PRICE ANALYSIS

Professional Power BI Dashboard Project Report

Project	Details
Project	Crude Oil Price Analysis Dashboard
Tool	Microsoft Power BI
Analysis scope	Brent and WTI price, spread, daily-change and period analysis

1. Executive Summary

The Crude Oil Price Analysis Dashboard is an interactive Power BI report developed to compare Brent and WTI crude-oil prices and related indicators. The dashboard combines KPI cards, historical trend analysis, monthly and yearly comparisons, daily price-change analysis, spread analysis, and a Year/Month slicer.

The dashboard is descriptive: it summarizes the information represented in the project data and shows how the measures change when the time filter is used.

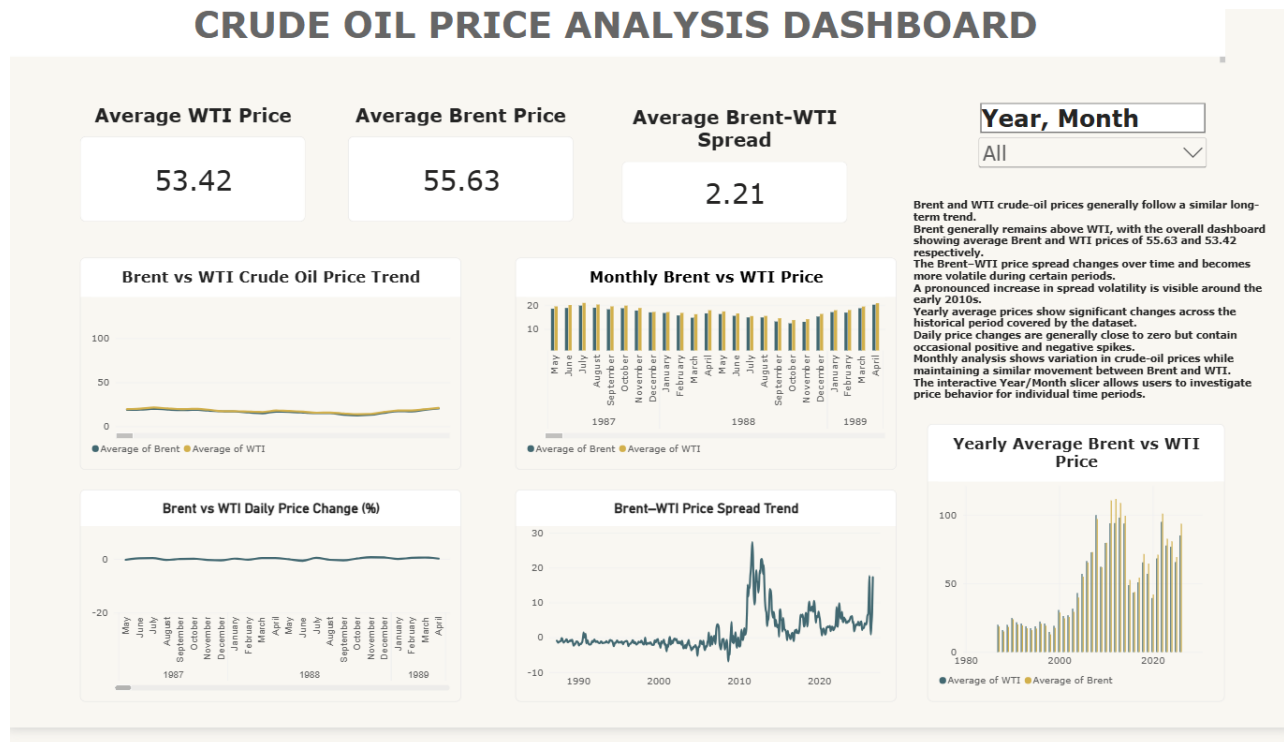


Figure 1. Final dashboard screenshot.

Unfiltered dashboard KPI values: Average WTI Price = 53.42; Average Brent Price = 55.63; Average Brent-WTI Spread = 2.21.

2. Project Objective

The objective is to build a clear and interactive visual analysis of Brent and WTI crude-oil prices, highlighting price levels, changes over time, and the relative price spread between the two benchmarks.

3. Dashboard Structure & KPI Framework

The final dashboard contains three KPI cards, five analytical visuals and a Year/Month slicer. Summary information is positioned above the detailed analytical views.

Dashboard Element	Purpose
Average WTI Price	Headline WTI price indicator.
Average Brent Price	Headline Brent price indicator.
Average Brent-WTI Spread	Headline relative-price indicator.
Brent vs WTI Crude Oil Price Trend	Long-term movement of the two benchmark prices.
Monthly Brent vs WTI Price	Month-level comparison in the selected time context.
Brent vs WTI Daily Price Change (%)	Positive and negative price-change movement.
Brent-WTI Price Spread Trend	Movement in the price difference between Brent and WTI.
Yearly Average Brent vs WTI Price	Year-to-year comparison of average benchmark prices.
Year/Month Slicer	Interactive filtering by time period.

4. KPI Summary

KPI	All-period value	Meaning
Average WTI Price	53.42	Average WTI value shown in the unfiltered dashboard.
Average Brent Price	55.63	Average Brent value shown in the unfiltered dashboard.
Average Brent-WTI Spread	2.21	Average difference between Brent and WTI shown in the unfiltered dashboard.

5. Interactive Filtering

The Year/Month slicer is configured as a compact dropdown. Selecting a year changes the current filter context, and the KPI cards and analytical visuals respond to the selection. The project testing performed during dashboard development confirmed that selecting a year changes the displayed measures while clearing the selection returns the dashboard to the overall view.

6. Dashboard Reading Sequence

- Start with the KPI cards for the current filter context.
- Use the main trend chart to compare long-term Brent and WTI movement.
- Use the monthly and yearly visuals for period-level comparisons.
- Use the daily price-change chart for short-term positive and negative movement.
- Use the spread trend to inspect changes in the relative Brent-WTI price difference.

7. Key Dashboard Observations

Brent and WTI co-movement

The trend views show Brent and WTI moving in broadly similar directions across the historical period represented in the dashboard.

Overall average relationship

In the unfiltered view, Average Brent is 55.63 and Average WTI is 53.42, while Average Brent-WTI Spread is 2.21.

Spread variability

The Brent-WTI spread changes across years rather than remaining constant. The spread visual shows both smaller and larger gaps.

Daily price-change behavior

The daily price-change visual is centered near the zero level for many periods, with selected periods showing larger positive or negative movements.

Yearly variation

The yearly comparison shows substantial variation in average Brent and WTI prices across the historical period.

Interactive filtering

The Year/Month slicer allows the same KPIs and charts to be inspected under a selected period.

8. Interpretation Notes

These observations are descriptive and are based only on the completed dashboard. The visuals show historical patterns and relationships in the displayed data, but they do not establish external causes for the observed movements.

The dashboard is intended for exploratory analysis and presentation. Forecasting and causal analysis are outside the scope of the completed page.

9. Project Scope and Reporting Discipline

The report uses only the information available in the completed dashboard and the supplied dashboard screenshot. Numerical values in this report are limited to the KPI values displayed in the unfiltered dashboard view. No external market statistics or explanatory claims have been added.

10. Conclusion

The completed Crude Oil Price Analysis Dashboard brings Brent and WTI crude-oil analysis into a single interactive Power BI view. The combination of KPI cards, historical trend, monthly comparison, daily price-change analysis, spread analysis, yearly comparison, and Year/Month filtering provides a practical structure for exploratory crude-oil price analysis.

11. Project Outcomes

- A consolidated interactive dashboard was created using the Brent and WTI measures available in the project.
- Three headline KPIs summarize WTI price, Brent price and the Brent-WTI spread.
- Five complementary visuals cover long-term trend, monthly comparison, daily change, spread movement and yearly comparison.
- The Year/Month slicer enables period-specific inspection without changing the underlying dashboard structure.
- The final page is organized for project demonstration and exploratory analysis.

12. Future Extension

Possible extensions compatible with the current dashboard include rolling averages, rolling volatility, selected-period versus all-period comparison measures, and a dedicated detail page. These are proposed future enhancements and are not part of the current dashboard.

13. Final Summary

The dashboard provides a compact visual framework for comparing Brent and WTI prices, understanding their spread, observing price changes, and interacting with the analysis through time-based filtering.

End of Report